TEKSTAC HANDS-ON

Spring MVC with Spring Boot

01) Sports Academy

Pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<parent>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-parent</artifactId>

<version>3.0.6</version>

<relativePath/> <!-- lookup parent from repository -->

</parent>

<groupId>com.example.demo</groupId>

<artifactId>SportsAcademy</artifactId>

<version>0.0.1-SNAPSHOT</version>

<name>SportsAcademy</name>

<description>SportsAcademy</description>

<properties>

<java.version>17</java.version>

</properties>

<dependencies>

<dependency>

<groupId>org.springframework.boot</groupId>
```

<artifactId>spring-boot-starter-web</artifactId>

</dependency>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-test</artifactId>

<scope>test</scope>

</dependency>

<dependency>

<groupId>org.springframework.restdocs</groupId>

<artifactId>spring-restdocs-mockmvc</artifactId>

<scope>test</scope>

</dependency>

<dependency>

<groupId>jakarta.validation</groupId>

<artifactId>jakarta.validation-api</artifactId>

</dependency>

<dependency>

<groupId>org.hibernate.validator</groupId>

<artifactId>hibernate-validator</artifactId>

</dependency>

<dependency>

<groupId>org.apache.tomcat.embed</groupId>

<artifactId>tomcat-embed-jasper</artifactId>

<scope>provided</scope>

</dependency>

</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-maven-plugin</artifactId>

</plugin>

</plugins>

</build>

</project>

InternationalizationConfig.java

```
package com.controller;


import java.util.Locale;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.web.servlet.LocaleResolver;
import org.springframework.web.servlet.config.annotation.InterceptorRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
import org.springframework.web.servlet.i18n.LocaleChangeInterceptor;
import org.springframework.web.servlet.i18n.SessionLocaleResolver;


@Configuration
public class InternationalizationConfig implements WebMvcConfigurer {


   @Bean
   public LocaleResolver localeResolver() {
      SessionLocaleResolver slr = new SessionLocaleResolver();
      slr.setDefaultLocale(Locale.ENGLISH);
      return slr;
   }
```

```
    @Bean
    public LocaleChangeInterceptor localeChangeInterceptor() {
        LocaleChangeInterceptor lci = new LocaleChangeInterceptor();
        lci.setParamName("language");
        return lci;
    }

    @Override
    public void addInterceptors(InterceptorRegistry registry) {
        registry.addInterceptor(localeChangeInterceptor());
    }
}
```

RegistrationController.java

```
package com.controller;

import java.util.ArrayList;
import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.ModelMap;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestMapping;

import com.model.Registration;
import com.service.RegistrationService;
```

```
@Controller
public class RegistrationController {

    @Autowired
    private RegistrationService service;

    @GetMapping("/showPage")
    public String showPage(@ModelAttribute("registration") Registration registration) {
        return "showPage";
    }

    @ModelAttribute("sportsList")
    public List<String> populateSports() {
        List<String> sports = new ArrayList<>();
        sports.add("Cricket");
        sports.add("Football");
        sports.add("Hockey");
        sports.add("Badminton");
        sports.add("Tennis");
        return sports;
    }

    @PostMapping("/registration")
    public String sportsRegistration(@ModelAttribute("registration") Registration registration,
ModelMap model) {
        int fees = service.sportsRegistration(registration);
        String message = "Thanks for visiting. Your monthly fees for " +
registration.getOptedSports() + " training is $" + fees;
        model.addAttribute("message", message);
        return "showPage";
```

```
    }
}
```

SportsAccademyApplication.java

```
package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.ComponentScan;

@SpringBootApplication
@ComponentScan(basePackages = {"com.controller", "com.service", "com.model"})
public class SportsAcademyApplication {

    public static void main(String[] args) {
        SpringApplication.run(SportsAcademyApplication.class, args);
    }
}
```

Registration.java

```
package com.model;

import org.springframework.stereotype.Component;

@Component
public class Registration {

    private String studentName;
    private int age;
    private String phoneNumber;
    private String optedSports;
```

```
public String getStudentName() {
    return studentName;
}

public void setStudentName(String studentName) {
    this.studentName = studentName;
}

public int getAge() {
    return age;
}

public void setAge(int age) {
    this.age = age;
}

public String getPhoneNumber() {
    return phoneNumber;
}

public void setPhoneNumber(String phoneNumber) {
    this.phoneNumber = phoneNumber;
}

public String getOptedSports() {
    return optedSports;
}
```

```
    public void setOptedSports(String optedSports) {
        this.optedSports = optedSports;
    }
}
```

RegistrationService.java

```
package com.service;

import org.springframework.stereotype.Service;
import com.model.Registration;

@Service
public class RegistrationService {

    public int sportsRegistration(Registration registration) {
        String sport = registration.getOptedSports();
        switch (sport) {
            case "Cricket": return 500;
            case "Football": return 350;
            case "Hockey": return 750;
            case "Badminton": return 400;
            case "Tennis": return 450;
            default: return 0;
        }
    }
}
```

ShowPage.jsp

```
<%@ taglib prefix="spring" uri="http://www.springframework.org/tags" %>
<%@ taglib prefix="form" uri="http://www.springframework.org/tags/form" %>
```

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1" isELIgnored="false"%>

<!DOCTYPE html>
<html>
<head>
<style>
#message {
    color: red;
}
</style>
</head>
<body style="background-color:lavender">

<!-- Language toggle links (do not modify) -->
<a href="/showPage?language=en">English</a>|<a
href="/showPage?language=fr">French</a>

<h1><center>Sports Academy</center></h1>

<form:form method="POST" action="/registration" modelAttribute="registration">
<table style="margin: 0px auto;">
<tr>
<td><spring:message code="label.studentName" /></td>
<td><form:input path="studentName" id="studentName" /></td>
</tr>
<tr>
<td><spring:message code="label.age" /></td>
<td><form:input path="age" id="age" /></td>
```

```
</tr>
<tr>
<td><spring:message code="label.phoneNumber" /></td>
<td><form:input path="phoneNumber" id="phoneNumber" /></td>
</tr>
<tr>
<td><spring:message code="label.optedSports" /></td>
<td>
<form:select path="optedSports" id="optedSports">
<form:options items="${sportsList}" />
</form:select>
</td>
</tr>
<tr>
<td colspan="2" align="center">
<input type="submit" name="submit" id="submit" value="Register" />
</td>
</tr>
<tr>
<td colspan="2">
<div id="message">${message}</div>
</td>
</tr>
</table>
</form:form>

</body>
</html>
```

02) Top Colleges List

Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>2.5.4</version> <!-- Update the version here -->
    <relativePath/> <!-- lookup parent from repository -->
  </parent>
  <groupId>com.example.demo</groupId>
  <artifactId>CollegeList</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name>CollegeList</name>
  <description>CollegeList</description>
  <properties>
    <java.version>17</java.version>
  </properties>
  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-test</artifactId>
```

```
      <scope>test</scope>
    </dependency>
    <dependency>
      <groupId>org.springframework.restdocs</groupId>
      <artifactId>spring-restdocs-mockmvc</artifactId>
      <scope>test</scope>
    </dependency>
    <dependency>
  <groupId>jakarta.validation</groupId>
  <artifactId>jakarta.validation-api</artifactId>
</dependency>
<dependency>
  <groupId>org.hibernate.validator</groupId>
  <artifactId>hibernate-validator</artifactId>
</dependency>
<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
<scope>provided</scope>
</dependency>
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>htmlunit-driver</artifactId>

</dependency>
                    <dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
```

```
</dependency>

<dependency>
<groupId>javax.servlet</groupId>
<artifactId>jstl</artifactId>
</dependency>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.12</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <build>
    <plugins>
      <plugin>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
      </plugin>
    </plugins>
  </build>
</project>
```

CollegeController.java

```
package com.controller;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.ModelMap;
```

```
import org.springframework.web.bind.annotation.*;
import org.springframework.web.servlet.ModelAndView;
import com.model.College;
import com.service.CollegeService;
import com.exception.UnavailableCollegeException;
import java.util.ArrayList;

@Controller
public class CollegeController {

   @Autowired
   private CollegeService service;

   @RequestMapping(value = "/showPage", method = RequestMethod.GET)
   public String showPage(@ModelAttribute("college") College college) {
      return "showPage";
   }

   @RequestMapping(value = "/viewCollegeList", method = RequestMethod.POST)
   public String searchCollege(@ModelAttribute("college") College college, ModelMap
model) {
      try {
         ArrayList<College> result = service.searchCollege(college);
         model.addAttribute("collegeList", result);
         return "viewCollegeList";
      } catch (UnavailableCollegeException e) {
         model.addAttribute("message", e.getMessage());
         return "exceptionPage";
      }
```

```
    }

    @ExceptionHandler(Exception.class)
    public ModelAndView exceptionHandler(Exception e) {
        ModelAndView mv = new ModelAndView("exceptionPage");
        mv.addObject("message", e.getMessage());
        return mv;
    }
}
```

CollegeListApplication.java

```
package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.ComponentScan;

// ✅ Scans all required packages for components like @Controller, @Service, etc.
@SpringBootApplication
@ComponentScan(basePackages = {"com.controller", "com.service", "com.model",
"com.exception"})
public class CollegesListApplication {

    public static void main(String[] args) {
        SpringApplication.run(CollegesListApplication.class, args);
    }
}
```

UnavailableCollegeException.java

```
package com.exception;

public class UnavailableCollegeException extends Exception {
    private static final long serialVersionUID = 1L;

    public UnavailableCollegeException(String msg) {
        super(msg); // ✅ Pass the message to the base Exception class
    }
}
```

College.java

```
package com.model;

import java.io.Serializable;

public class College implements Serializable {

    private String collegeName;
    private String location;
    private String offeringCourse;
    private double annualFees;

    public String getCollegeName() {
        return collegeName;
    }

    public void setCollegeName(String collegeName) {
        this.collegeName = collegeName;
```

```
    }

    public String getLocation() {
        return location;
    }

    public void setLocation(String location) {
        this.location = location;
    }

    public String getOfferingCourse() {
        return offeringCourse;
    }

    public void setOfferingCourse(String offeringCourse) {
        this.offeringCourse = offeringCourse;
    }

    public double getAnnualFees() {
        return annualFees;
    }

    public void setAnnualFees(double annualFees) {
        this.annualFees = annualFees;
    }

    public College(String collegeName, String location, String offeringCourse, double
annualFees) {
        this.collegeName = collegeName;
```

```
        this.location = location;
        this.offeringCourse = offeringCourse;
        this.annualFees = annualFees;
    }

    public College() {
    }
}
```

CollegeService.java

```
package com.service;

import java.util.ArrayList;
import org.springframework.stereotype.Service;
import com.model.College;
import com.exception.UnavailableCollegeException;

@Service // ✅ Configures this class as a Spring-managed service component
public class CollegeService {

    ArrayList<College> collegeList = new ArrayList<College>();

    // ✅ Search the collegeList based on offeringCourse and annualFees
    public ArrayList<College> searchCollege(College college) throws
UnavailableCollegeException {
        ArrayList<College> result = new ArrayList<>();
        String inputCourse = college.getOfferingCourse().toLowerCase();
        double inputFees = college.getAnnualFees();
```

```
        for (College c : collegeList) {
            if (c.getOfferingCourse().equalsIgnoreCase(inputCourse) && c.getAnnualFees() <=
inputFees) {
                result.add(c);
            }
        }

        if (result.isEmpty()) {
            throw new UnavailableCollegeException("Sorry!! No colleges are available for the
course " +
                    college.getOfferingCourse() + " with in the given fees range");
        }

        return result;
    }

    // Do not modify or delete the below code
    public CollegeService() {
        addColleges();
    }

    public void addColleges() {
        collegeList.add(new College("IIT Kanpur", "Kanpur", "Engineering", 10000));
        collegeList.add(new College("HCS Womens college ", "Delhi", "BCA", 15000));
        collegeList.add(new College("Vidya", "Kerala", "Engineering", 65000));
        collegeList.add(new College("IIM", "Ahmedabad", "MBA", 85000));
        collegeList.add(new College("MTET", "Coimbatore", "BArch", 100000));
        collegeList.add(new College("CET", "Kerala", "Engineering", 20000));
        collegeList.add(new College("Christ", "Bangalore", "Bcom", 50000));
        collegeList.add(new College("AIIMS", "Delhi", "MBBS", 200000));
```

```
        collegeList.add(new College("Loyala", "Chennai", "Bcom", 25000));
        collegeList.add(new College("Symbiosis Law School", "Pune", "BArch", 150000));
    }
}
```

Application.properties

```
server.port=9088
spring.mvc.view.prefix = /WEB-INF/views/
spring.mvc.view.suffix = .jsp
spring.mvc.static-path-pattern=/resources/**
```

ExceptionPage.jsp

```
<%@page isELIgnored="false" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/functions" prefix="fn" %>
<html>
<body bgcolor="lavender">
<h1><center> Top Colleges List </center></h1>
<center>

<table border="1" cellpadding="10">
<tr>
<th>College Name</th>
<th>Location</th>
<th>Offering Course</th>
<th>Annual Fees</th>
</tr>
<c:forEach var="college" items="${collegeList}">
<tr>
```

```
<td>${college.collegeName}</td>
<td>${college.location}</td>
<td>${college.offeringCourse}</td>
<td>${college.annualFees}</td>
</tr>
</c:forEach>
</table>

<div>
<center><a href="showPage">Back to search</a></center>
</div>

</center>
</body>
</html>
```

showPage.jsp

```
<%@ taglib prefix="spring" uri="http://www.springframework.org/tags" %>
<%@ taglib prefix="sf" uri="http://www.springframework.org/tags/form" %>
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form"%>
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1" isELIgnored="false"%>

<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<body style="background-color:lavender">
<h1><center> Top Colleges List</center></h1>
```

```
<center>
<form:form method="POST" action="viewCollegeList" modelAttribute="college">
<table>
<tr>
<td>Course:</td>
<td><form:input path="offeringCourse" id="offeringCourse"/></td>
</tr>
<tr>
<td>Annual Fees:</td>
<td><form:input path="annualFees"/></td>
</tr>
<tr>
<td colspan="2" align="center">
<input type="submit" value="Search" id="submit"/>
</td>
</tr>
</table>
</form:form>
</center>

</body>
</html>
```

viewCollegeList.jsp

```
<%@ taglib prefix="spring" uri="http://www.springframework.org/tags" %>
<%@ taglib prefix="sf" uri="http://www.springframework.org/tags/form" %>
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form"%>
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1" isELIgnored="false"%>
```

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<body style="background-color:lavender">
<h1><center> Top Colleges List</center></h1>

<center>
<form:form method="POST" action="viewCollegeList" modelAttribute="college">
<table>
<tr>
<td>Course:</td>
<td><form:input path="offeringCourse" id="offeringCourse"/></td>
</tr>
<tr>
<td>Annual Fees:</td>
<td><form:input path="annualFees"/></td>
</tr>
<tr>
<td colspan="2" align="center">
<input type="submit" value="Search" id="submit"/>
</td>
</tr>
</table>
</form:form>
</center>

</body>
</html>
```

03) PulsePursuit Tournament

Pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"


        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<parent>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-parent</artifactId>

<version>3.0.6</version>

<relativePath/> <!-- lookup parent from repository -->

</parent>

<groupId>com.example.demo</groupId>

<artifactId>PulsePursuit</artifactId>

<version>0.0.1-SNAPSHOT</version>

<name>PulsePursuit</name>

<description>PulsePursuit</description>

<properties>

<java.version>17</java.version>

</properties>

<dependencies>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-web</artifactId>

</dependency>
```

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-test</artifactId>
<scope>test</scope>
</dependency>
<dependency>
<groupId>org.springframework.restdocs</groupId>
<artifactId>spring-restdocs-mockmvc</artifactId>
<scope>test</scope>
</dependency>
<dependency>
<groupId>jakarta.validation</groupId>
<artifactId>jakarta.validation-api</artifactId>
</dependency>
<dependency>
<groupId>org.hibernate.validator</groupId>
<artifactId>hibernate-validator</artifactId>
</dependency>
<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
<scope>provided</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>

```
</plugin>
</plugins>
</build>
</project>
```

EventController.java

```
package com.controller;

import java.util.ArrayList;
import java.util.List;

import jakarta.validation.Valid;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.ModelMap;
import org.springframework.validation.BindingResult;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestMethod;

import com.model.Registration;
import com.service.EventService;

@Controller
public class EventController {
    @Autowired
    private EventService service;
    @ModelAttribute("eventTypeList")
```

```
    public List<String> populateEventType(){
        List<String> events = new ArrayList<>();
        events.add("GolfTournament");
        events.add("Triathlon");
        events.add("Marathon");
        events.add("HikingChallenge");
        return events;
    }
    @RequestMapping(value="/showPage", method=RequestMethod.GET)
    public String showPage(@ModelAttribute("registration") Registration registration) {
        return "showPage";
    }
    @RequestMapping(value="/register", method=RequestMethod.POST)
    public String registerEvent(@ModelAttribute("registration") @Valid Registration
registration,
                                BindingResult result,
                                ModelMap model) {
        if (result.hasErrors()) {
            return "showPage";
        }
        double cost = service.calculateRegistrationCost(registration);
        model.addAttribute("message", "Dear " + registration.getParticipantName() + ", You
have to pay Rs." + cost);
        return "showPage";
    }
}
```

EventRegistrationApplication.java

```
package com.example.demo;
```

```
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.ComponentScan;

@SpringBootApplication
@ComponentScan({"com.controller","com.model","com.service"})
public class EventRegistrationApplication {
	public static void main(String[] args) {
		SpringApplication.run(EventRegistrationApplication.class, args);
	}

}
```

Registration.java

```
package com.model;

import jakarta.validation.constraints.*;

public class Registration {
   @NotBlank(message = "Participant Name cannot be empty")
   private String participantName;
   @Pattern(regexp = "^[6-9][0-9]{9}$", message = "Phone number should have 10 digits and first digit should be 6,7 ,8 or 9")
   private String phoneNumber;
   @NotBlank(message = "Email cannot be empty")
   @Email(message = "Invalid email format")
   private String email;
   @Min(value = 18, message = "Age must be greater than 18")
   @Max(value = 25, message = "Age must be smaller than 25")
```

```
private int age;

private String eventType;
public String getParticipantName() {
    return participantName;
}

public void setParticipantName(String participantName) {
    this.participantName = participantName;
}

public String getPhoneNumber() {
    return phoneNumber;
}

public void setPhoneNumber(String phoneNumber) {
    this.phoneNumber = phoneNumber;
}

public String getEmail() {
    return email;
}

public void setEmail(String email) {
    this.email = email;
}

public int getAge() {
    return age;
```

```
    }

    public void setAge(int age) {
        this.age = age;
    }
    public String getEventType() {
        return eventType;
    }

    public void setEventType(String eventType) {
        this.eventType = eventType;
    }
}
```

EventService.java

```
package com.service;

import org.springframework.stereotype.Service;

import com.model.Registration;

@Service
public class EventService {
    public double calculateRegistrationCost(Registration registration) {
        double regCost = 0.0;
        switch (registration.getEventType()) {
            case "GolfTournament":
                regCost = 2000;
                break;
```

```
            case "Triathlon":
                regCost = 1000;
                break;
            case "Marathon":
                regCost = 1200;
                break;
            case "HikingChallenge":
                regCost = 1500;
                break;
        }
        return regCost + (regCost * 0.18);
    }
}
```

Application.properties

```
#Do not change the configurations
#Value for server.port can be changed
#include other needed configurations
server.port=9095
spring.mvc.view.prefix = /WEB-INF/views/
spring.mvc.view.suffix = .jsp
spring.mvc.static-path-pattern=/resources/**
```

showPage.jsp

```
<%@ taglib prefix="form" uri="http://www.springframework.org/tags/form" %>
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"

    pageEncoding="ISO-8859-1" isELIgnored="false"%>
```

```
<html>
<body style="background-color:lavender">
<h1><center> Welcome to PulsePursuit Tournament</center></h1>

<center>
<form:form modelAttribute="registration" action="register" method="post">
<table>
<tr>
<td>Participant Name</td>
<td>
<form:input path="participantName" id="participantName"/>
<form:errors path="participantName" cssStyle="color:red"/>
</td>
</tr>
<tr>
<td>Phone Number</td>
<td>
<form:input path="phoneNumber" id="phoneNumber"/>
<form:errors path="phoneNumber" cssStyle="color:red"/>
</td>
</tr>
<tr>
<td>Email</td>
<td>
<form:input path="email" id="email"/>
<form:errors path="email" cssStyle="color:red"/>
</td>
</tr>
<tr>
```

```
<td>Age</td>
<td>
<form:input path="age" id="age"/>
<form:errors path="age" cssStyle="color:red"/>
</td>
</tr>
<tr>
<td>Event Type</td>
<td>
<form:select path="eventType" id="eventType">
<form:options items="${eventTypeList}"/>
</form:select>
</td>
</tr>
<tr>
<td colspan="2" align="center">
<input type="submit" id="submit" value="Register"/>
</td>
</tr>
</table>
</form:form>

<c:if test="${not empty message}">
<h3 style="color:green">${message}</h3>
</c:if>
</center>

</body>
</html>
```

04) Dexler Preschool

Pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">

        <modelVersion>4.0.0</modelVersion>

        <parent>

                <groupId>org.springframework.boot</groupId>

                <artifactId>spring-boot-starter-parent</artifactId>

                <version>3.0.6</version>

                <relativePath/> <!-- lookup parent from repository -->

        </parent>

  <groupId>com.example.demo</groupId>

  <artifactId>Dexler</artifactId>

  <version>0.0.1-SNAPSHOT</version>

  <name>DexlerPreschool</name>

  <description>DexlerPreschool</description>

  <properties>

    <java.version>17</java.version>

  </properties>

  <dependencies>

    <dependency>

      <groupId>org.springframework.boot</groupId>

      <artifactId>spring-boot-starter-web</artifactId>

    </dependency>

    <dependency>
```

```
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-test</artifactId>
      <scope>test</scope>
    </dependency>
    <dependency>
      <groupId>org.springframework.restdocs</groupId>
      <artifactId>spring-restdocs-mockmvc</artifactId>
      <scope>test</scope>
    </dependency>
    <dependency>
  <groupId>jakarta.validation</groupId>
  <artifactId>jakarta.validation-api</artifactId>
</dependency>
<dependency>
  <groupId>org.hibernate.validator</groupId>
  <artifactId>hibernate-validator</artifactId>
</dependency>
<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
<scope>provided</scope>
</dependency>

  </dependencies>
  <build>
    <plugins>
      <plugin>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
```

```
    </plugin>
   </plugins>
  </build>
</project>
```

AdmissionController.java

```
package com.controller;

import java.util.ArrayList;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.ModelMap;
import org.springframework.validation.BindingResult;
import org.springframework.web.bind.annotation.*;

import com.model.Admission;
import com.service.AdmissionService;
import com.validate.CustomValidator;

import jakarta.validation.Valid;

@Controller
public class AdmissionController {

   @Autowired
   private CustomValidator custValidator;

   @Autowired
```

```
private AdmissionService admissionService;

@RequestMapping(value = "/showPage", method = RequestMethod.GET)
public String showPage(@ModelAttribute("admission") Admission admission) {
   return "showPage";
}

@ModelAttribute("programNames")
public ArrayList<String> populateAvailablePrograms() {
   ArrayList<String> programs = new ArrayList<>();
   programs.add("Play Group");
   programs.add("Nursery");
   programs.add("Euro Junior");
   programs.add("Euro Senior");
   return programs;
}

@RequestMapping(value = "/register", method = RequestMethod.POST)
public String registerPreschooler(
      @Valid @ModelAttribute("admission") Admission admission,
      BindingResult result,
      ModelMap model) {

   custValidator.validate(admission, result);

   if (result.hasErrors()) {
      return "showPage";
   }
```

```
        double fee = admissionService.calculateProgramFee(admission);
        model.addAttribute("message", "Preschool registration successful for " +
admission.getChildName() + ", Program Fees: " + fee);
        return "success";
    }
}
```

PreSchoolApllication.java

```
package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.ComponentScan;

@SpringBootApplication
@ComponentScan({"com.controller","com.model","com.service","com.validate"})
public class PreschoolApplication {

        public static void main(String[] args) {
                SpringApplication.run(PreschoolApplication.class, args);
        }

}
```

Admission.java

```
package com.model;

import jakarta.validation.constraints.*;

public class Admission {
```

```
@NotBlank(message = "Child name should not be blank")
private String childName;

@NotBlank(message = "Contact Number should not be blank")
private String contactNumber;

@Min(value = 2, message = "Age must be greater than or equal to 2")
@Max(value = 7, message = "Age must be lesser than or equal to 7")
private int age;

@NotBlank(message = "Email id should not be blank")
@Email(message = "Email format is not valid")
private String emailId;

private String programName;

// Getters and Setters
public String getChildName() {
   return childName;
}

public void setChildName(String childName) {
   this.childName = childName;
}

public String getContactNumber() {
   return contactNumber;
}
```

```
    public void setContactNumber(String contactNumber) {
        this.contactNumber = contactNumber;
    }

    public int getAge() {
        return age;
    }

    public void setAge(int age) {
        this.age = age;
    }

    public String getEmailId() {
        return emailId;
    }

    public void setEmailId(String emailId) {
        this.emailId = emailId;
    }

    public String getProgramName() {
        return programName;
    }

    public void setProgramName(String programName) {
        this.programName = programName;
    }
}
```

AdmissionService.java

```
package com.service;

import org.springframework.stereotype.Service;
import com.model.Admission;

@Service
public class AdmissionService {

    public double calculateProgramFee(Admission admission) {
        String program = admission.getProgramName();

        if (program == null) return 0;

        switch (program) {
            case "Play Group":
                return 600;
            case "Nursery":
                return 800;
            case "Euro Junior":
                return 1000;
            case "Euro Senior":
                return 1200;
            default:
                return 0;
        }
    }
}
```

CustomValidator.java

```
package com.validate;


import com.model.Admission;
import org.springframework.stereotype.Component;
import org.springframework.validation.Errors;
import org.springframework.validation.Validator;


@Component
public class CustomValidator implements Validator {


    @Override
    public boolean supports(Class<?> clazz) {
        return Admission.class.isAssignableFrom(clazz);
    }


    @Override
    public void validate(Object target, Errors errors) {
        Admission admission = (Admission) target;


        String name = admission.getChildName();
        if (name != null && !name.matches("[a-zA-Z ]{2,}")) {
            errors.rejectValue("childName", "", "Name should contain only alphabets and spaces with a minimum of 2 characters");
        }


        String contact = admission.getContactNumber();
        if (contact != null && !contact.matches("^[6-9]\\d{9}$")) {
            errors.rejectValue("contactNumber", "", "Contact Number should be of 10 digits and start with a number between 6 and 9");
```

```
    }
  }
}
```

Application.properties

```
server.port=9088
spring.mvc.view.prefix = /WEB-INF/views/
spring.mvc.view.suffix = .jsp
spring.mvc.static-path-pattern=/resources/**
```

showPage.jsp

```
<%@ taglib prefix="spring" uri="http://www.springframework.org/tags" %>
<%@ taglib prefix="sf" uri="http://www.springframework.org/tags/form" %>
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form"%>
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
   pageEncoding="ISO-8859-1" isELIgnored="false"%>

<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<style>
#message {
   color: brown;
}
</style>
</head>
<body style="background-color:lavender">
```

```
<h1><center>Preschool Enrollment</center></h1>

<form:form modelAttribute="admission" action="register" method="post">
<table style="margin: 0px auto; margin-left: auto; margin-right:auto">
<tr>
<td>Child Name:</td>
<td><form:input path="childName" id="childName"/></td>
<td><form:errors path="childName" cssClass="error"/></td>
</tr>
<tr>
<td>Contact Number:</td>
<td><form:input path="contactNumber" id="contactNumber"/></td>
<td><form:errors path="contactNumber" cssClass="error"/></td>
</tr>
<tr>
<td>Age:</td>
<td><form:input path="age" id="age"/></td>
<td><form:errors path="age" cssClass="error"/></td>
</tr>
<tr>
<td>Email Id:</td>
<td><form:input path="emailId" id="emailId"/></td>
<td><form:errors path="emailId" cssClass="error"/></td>
</tr>
<tr>
<td>Program Name:</td>
<td>
<form:select path="programName" id="programName">
<form:options items="${programNames}"/>
```

```
</form:select>
</td>
</tr>
<tr>
<td colspan="2" align="center">
<input type="submit" value="Register" id="submit"/>
</td>
</tr>
</table>
</form:form>

</body>
</html>
```

Success.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"

    pageEncoding="ISO-8859-1"%>
<%@page isELIgnored="false" %>

<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Preschool Registration Success</title>
</head>
<body style="background-color:lavender">
<h1><center>Dexler Preschool</center></h1>
<center>
```

<h2>${message}</h2>

Home

</body>

</html>

05) SignUp Form

Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.4</version> <!-- Update the version here -->
<relativePath/> <!-- lookup parent from repository -->
</parent>
<groupId>com.example.demo</groupId>
<artifactId>SignUpForm</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>SignUpForm</name>
<description>SignUpForm</description>
<properties>
<java.version>17</java.version>
</properties>
<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>
<dependency>
<groupId>org.springframework.boot</groupId>
```

<artifactId>spring-boot-starter-test</artifactId>

<scope>test</scope>

</dependency>

<dependency>

<groupId>org.springframework.restdocs</groupId>

<artifactId>spring-restdocs-mockmvc</artifactId>

<scope>test</scope>

</dependency>

<dependency>

<groupId>jakarta.validation</groupId>

<artifactId>jakarta.validation-api</artifactId>

</dependency>

<dependency>

<groupId>org.hibernate.validator</groupId>

<artifactId>hibernate-validator</artifactId>

</dependency>

<dependency>

<groupId>org.apache.tomcat.embed</groupId>

<artifactId>tomcat-embed-jasper</artifactId>

<scope>provided</scope>

</dependency>

<dependency>

<groupId>org.seleniumhq.selenium</groupId>

<artifactId>htmlunit-driver</artifactId>

</dependency>

<dependency>

<groupId>org.seleniumhq.selenium</groupId>

<artifactId>selenium-java</artifactId>

</dependency>

```
<dependency>
<groupId>javax.servlet</groupId>
<artifactId>jstl</artifactId>
</dependency>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>4.12</version>
<scope>test</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```

AccountController.java

```
package com.controller;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.ModelMap;
import org.springframework.web.bind.annotation.GetMapping;
```

```
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.PostMapping;

import com.model.Account;
import com.service.AccountService;

@Controller
public class AccountController {

    @Autowired
    private AccountService service;

    @GetMapping("/showPage")
    public String showPage(@ModelAttribute("account") Account account) {
        return "showPage";
    }

    @PostMapping("/signUp")
    public String signup(@ModelAttribute("account") Account account, ModelMap model) {
        boolean valid = service.signup(account);
        if (valid) {
            model.addAttribute("message", "Your Account is Created");
        } else {
            model.addAttribute("message", "Invalid Password");
        }
        return "showPage";
    }
}
```

InternationalizationConfig.java

```
package com.controller;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.web.servlet.LocaleResolver;
import org.springframework.web.servlet.config.annotation.InterceptorRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
import org.springframework.web.servlet.i18n.LocaleChangeInterceptor;
import org.springframework.web.servlet.i18n.SessionLocaleResolver;

import java.util.Locale;

@Configuration
public class InternationalizationConfig implements WebMvcConfigurer {

    @Bean
    public LocaleResolver localeResolver() {
        SessionLocaleResolver slr = new SessionLocaleResolver();
        slr.setDefaultLocale(Locale.ENGLISH);
        return slr;
    }

    @Bean
    public LocaleChangeInterceptor localeChangeInterceptor() {
        LocaleChangeInterceptor lci = new LocaleChangeInterceptor();
        lci.setParamName("language");
        return lci;
    }
```

```
    @Override
    public void addInterceptors(InterceptorRegistry registry) {
        registry.addInterceptor(localeChangeInterceptor());
    }
}
```

SingupFormApplication.java

```
package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.ComponentScan;

@SpringBootApplication
@ComponentScan({"com.controller","com.model","com.service"})
public class SignupFormApplication {
    public static void main(String[] args) {
        SpringApplication.run(SignupFormApplication.class, args);
    }
}
```

Account.java

```
package com.model;

import org.springframework.stereotype.Component;

@Component
public class Account {
```

```
    private String userName;
    private String gender;
    private String phoneNumber;
    private String email;
    private String password;

    public String getUserName() { return userName; }
    public void setUserName(String userName) { this.userName = userName; }

    public String getGender() { return gender; }
    public void setGender(String gender) { this.gender = gender; }

    public String getPhoneNumber() { return phoneNumber; }
    public void setPhoneNumber(String phoneNumber) { this.phoneNumber = phoneNumber;
}

    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }

    public String getPassword() { return password; }
    public void setPassword(String password) { this.password = password; }
}
```

AccountService.java

```
package com.service;

import org.springframework.stereotype.Service;
import com.model.Account;
```

```
@Service
public class AccountService {

    public boolean signup(Account account) {
        String password = account.getPassword();
        if (password == null) return false;

        boolean hasLetter = password.matches(".*[A-Za-z].*");
        boolean hasDigit = password.matches(".*\\d.*");
        boolean hasSpecial = password.matches(".*[^A-Za-z0-9].*");

        return hasLetter && hasDigit && hasSpecial;
    }
}
```

Application.properties

```
server.port=9088
spring.mvc.view.prefix = /WEB-INF/views/
spring.mvc.view.suffix = .jsp
spring.mvc.static-path-pattern=/resources/**
spring.resources.static-locations=classpath:/static/
```

message.properties

```
label.userName=Enter the User Name
label.gender=Gender
label.male=Male
label.female=Female
label.phoneNumber=Enter the Phone Number
label.email=Enter the Email Address
```

label.password=Create a Password

message_properties.fr

label.userName=Enter the User Name in French

label.gender=Gender

label.male=Male

label.female=Female

label.phoneNumber=Enter the Phone Number in French

label.email=Enter the Email Address in French

label.password=Create a Password in French

style.css

```
@import
url('https://fonts.googleapis.com/css?family=Poppins:400,500,600,700&display=swap');
*{
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: 'Poppins', sans-serif;
}

/* Style the body */
html,body{
  display: grid;
  height: 100%;
  width: 100%;
  place-items: center;
  background: #f2f2f2;
  background: linear-gradient(175deg, #c850c0, #21c8d4);
```

```
}

/* Style selected text */
::selection{
  background: #4158d0;
  color: #fff;
}

/* Style the wrapper container */
.wrapper{
  width: 380px;
  background: #fff;
  border-radius: 15px;
  box-shadow: 0px 15px 20px rgba(0,0,0,0.1);
}

/* Style the title */
.wrapper .title{
  font-size: 35px;
  font-weight: 600;
  text-align: center;
  line-height: 100px;
  color: #fff;
  user-select: none;
  border-radius: 15px 15px 0 0;
  background: linear-gradient(-135deg, #c850c0, #21c8d4);
}

/* Style the form container */
```

```
.wrapper form{
  padding: 10px 30px 50px 30px;
}

/* Style form fields and labels */
.wrapper form .field{
  height: 50px;
  width: 100%;
  margin-top: 20px;
  position: relative;
}

.wrapper form .field input{
  height: 100%;
  width: 100%;
  outline: none;
  font-size: 17px;
  padding-left: 20px;
  border: 1px solid lightgrey;
  border-radius: 25px;
  transition: all 0.3s ease;
}

.wrapper form .field input:focus,
form .field input:valid{
  border-color: #4158d0;
}

.wrapper form .field label{
```

```
  position: absolute;
  top: 50%;
  left: 20px;
  color: #999999;
  font-weight: 400;
  font-size: 17px;
  pointer-events: none;
  transform: translateY(-50%);
  transition: all 0.3s ease;
}

form .field input:focus ~ label,
form .field input:valid ~ label{
  top: 0%;
  font-size: 16px;
  color: #4158d0;
  background: #fff;
  transform: translateY(-50%);
  font-weight: bold; /* Add this line to bold the label */
}

/* Style the content section */
form .content{
  display: flex;
  width: 100%;
  height: 50px;
  font-size: 16px;
  align-items: center;
  justify-content: space-around;
```

```
}

/* Style checkboxes */
form .content .checkbox{
  display: flex;
  align-items: center;
  justify-content: center;
}

form .content input{
  width: 15px;
  height: 15px;
  background: red;
}

form .content label{
  color: #262626;
  user-select: none;
  padding-left: 5px;
}

form .content .pass-link{
  color: "";
}

/* Style the submit button */
form .field input[type="submit"]{
  color: #fff;
  border: none;
```

```
  padding-left: 0;
  margin-top: 0px;
  font-size: 20px;
  font-weight: 500;
  cursor: pointer;
  background: linear-gradient(-135deg, #c850c0, #21c8d4);
  transition: all 0.3s ease;
}

form .field input[type="submit"]:active{
  transform: scale(0.95);
}

/* Style the signup link */
form .signup-link{
  color: #262626;
  margin-top: 20px;
  text-align: center;
}

/* Style links within the form */
form .pass-link a,
form .signup-link a{
  color: #4158d0;
  text-decoration: none;
}

/* Style hover state for links */
form .pass-link a:hover,
```

```
form .signup-link a:hover{

  text-decoration: underline;
}

#result {
  color: #fff;
  font-size: 40px;
}

/* Add this CSS to style the radio buttons in the same line */
.wrapper form .gender-field {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-top: 20px;
}

.wrapper form .gender-field .radio-group {
  display: flex;
  align-items: center;
}

.wrapper form .gender-field input[type="radio"] {
  appearance: none;
  -webkit-appearance: none;
  width: 20px;
  height: 20px;
  border-radius: 50%;
```

```
  border: 2px solid #21c8d4;
  cursor: pointer;
  transition: background-color 0.3s ease, border-color 0.3s ease;
}

/* Style the radio buttons when checked */
.wrapper form .gender-field input[type="radio"]:checked {
  background-color: #c850c0;
  border-color: #c850c0;
}

/* Style the labels of the checked radio buttons */
.wrapper form .gender-field input[type="radio"] + label {
  color: #262626;
  font-size: 17px;
  cursor: pointer;
  margin-left: 5px; /* Add spacing between radio button and label */
  transition: color 0.3s ease;
}

/* Style the labels of the checked radio buttons */
.wrapper form .gender-field input[type="radio"]:checked + label {
  color: #4158d0;
  font-weight: 600;
}
```

Showpage.jsp

```
<%@ taglib prefix="spring" uri="http://www.springframework.org/tags" %>
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form"%>
```

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1" isELIgnored="false"%>

<html>
<head>
<meta charset="utf-8">
<title>Sign Form</title>
<link rel="stylesheet" type="text/css" href="/resources/css/style.css"/>
</head>
<body>
<p><a href="/showPage?language=en">English</a> | <a
href="/showPage?language=fr">French</a></p>
<div id="result"></div>
<div class="wrapper">
<div class="title">Create an Account</div>
<form:form action="signUp" method="post" modelAttribute="account">

        <div class="field">
<form:input path="userName" id="userName" required="true"/>
<label for="userName"><spring:message code="label.userName"/></label>
</div>

        <div class="gender-field">
<label><spring:message code="label.gender"/></label>
<div class="radio-group">
<form:radiobutton path="gender" id="male" value="Male"/>
<label for="male"><spring:message code="label.male"/></label>
</div>
<div class="radio-group">
```

```
<form:radiobutton path="gender" id="female" value="Female"/>
<label for="female"><spring:message code="label.female"/></label>
</div>
</div>

        <div class="field">
<form:input path="phoneNumber" id="phoneNumber" required="true"/>
<label for="phoneNumber"><spring:message code="label.phoneNumber"/></label>
</div>

        <!-- FIXED EMAIL FIELD -->
<div class="field">
<form:input path="email" id="email" type="email" required="true"/>
<label for="email"><spring:message code="label.email"/></label>
</div>

        <div class="field">
<form:password path="password" id="password" required="true"/>
<label for="password"><spring:message code="label.password"/></label>
</div>

        <div class="field">
<input type="submit" value="Sign" id="submit"/>
</div>

        <div class="field">
<label id="message">${message}</label>
</div>
```

```
    </form:form>
</div>
</body>
</html>
```

6) Swift Cart Portal

Pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>3.0.6</version>
<relativePath /> <!-- lookup parent from repository -->
</parent>
<groupId>com.spring.app</groupId>
<artifactId>SwiftCartPortal</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>SwiftCartPortal</name>
<description>SwiftCartPortal</description>
<properties>
<java.version>17</java.version>
</properties>
<dependencies>
```

```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-test</artifactId>
<scope>test</scope>
</dependency>
<dependency>
<groupId>org.springframework.restdocs</groupId>
<artifactId>spring-restdocs-mockmvc</artifactId>
<scope>test</scope>
</dependency>
<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
<scope>provided</scope>
</dependency>
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>htmlunit-driver</artifactId>
</dependency>
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
</dependency>

                <dependency>
```

<groupId>jakarta.servlet</groupId>

<artifactId>jakarta.servlet-api</artifactId>

<scope>provided</scope>

</dependency>

<dependency>

<groupId>javax.servlet</groupId>

<artifactId>jstl</artifactId>

<version>1.2</version>

</dependency>

<dependency>

<groupId>junit</groupId>

<artifactId>junit</artifactId>

<version>4.12</version>

<scope>test</scope>

</dependency>

</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-maven-plugin</artifactId>

</plugin>

</plugins>

</build>

</project>

CustomerControl.java

package com.controller;

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestMethod;

import com.model.Customer;
import com.service.CustomerService;

@Controller  // ✅ Marks this class as a Spring MVC controller
public class CustomerController {

    @Autowired  // ✅ Injects the CustomerService bean
    private CustomerService service;

    @RequestMapping(value = "/index", method = RequestMethod.GET)
    public String registrationPage(@ModelAttribute("customer") Customer customer) {
        return "index";  // ✅ Loads the registration form
    }

    @RequestMapping(value = "/registerCustomer", method = RequestMethod.POST)
    public String validateInfo(@ModelAttribute("customer") Customer customer) {
        int result = service.validateInfo(customer);

        if (result == 1) {
            return "success";  // ✅ Valid registration
        } else if (result == -2) {
            return "invalidEmail";  // ✅ Invalid email
        } else {
```

```
            return "invalidMobileNumber";  // ✅ Invalid mobile number
        }
    }
}
```

SwiftCartPoralApplication.java

```
package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.ComponentScan;

@SpringBootApplication
@ComponentScan(basePackages = {"com.controller", "com.service", "com.model"})
public class SwiftCartPortalApplication {

    public static void main(String[] args) {
        SpringApplication.run(SwiftCartPortalApplication.class, args);
    }
}
```

Customer.java

```
package com.model;



public class Customer {
            private String name;
            private String emaillId;
            private long mobileNumber;
```

```
	public String getName() {
		return name;
	}
	public void setName(String name) {
		this.name = name;
	}
	public String getEmailId() {
		return emailId;
	}
	public void setEmailId(String emailId) {
		this.emailId = emailId;
	}
	public long getMobileNumber() {
		return mobileNumber;
	}
	public void setMobileNumber(long mobileNumber) {
		this.mobileNumber = mobileNumber;
	}


}
```

CustomerService.java

```
package com.service;

import com.model.Customer;
import org.springframework.stereotype.Service;

@Service  // ✅ Marks this class as a Spring-managed service
```

```
public class CustomerService {

    public int validateInfo(Customer customer) {
        String email = customer.getEmailId();
        long mobile = customer.getMobileNumber();

        // ✅ Validate email format
        boolean validEmail = email != null && email.matches("^[\\w.-]+@[\\w.-]+\\.\\w{2,}$");

        // ✅ Validate mobile number: must be 10 digits starting with 6, 7, 8, or 9
        boolean validMobile = String.valueOf(mobile).matches("^[6-9]\\d{9}$");

        if (!validEmail) return -2;
        if (!validMobile) return -1;
        return 1;
    }
}
```

Application.properties

```
server.port=9088
spring.mvc.view.prefix = /WEB-INF/views/
spring.mvc.view.suffix = .jsp
spring.mvc.static-path-pattern=/resources/**
```

index.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
pageEncoding="ISO-8859-1"%>
<%@ taglib prefix="sf" uri="http://www.springframework.org/tags/form"%>
<%@ taglib prefix="s" uri="http://www.springframework.org/tags"%>
```

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<body bgcolor="lightgreen">
<center>
<h1>Swift Cart Portal</h1>
<sf:form method="POST" action="/registerCustomer" modelAttribute="customer">
<table>
<tr>
<td>Name:</td>
<td><sf:input path="name" id="name"/></td>
</tr>
<tr>
<td>Email ID:</td>
<td><sf:input path="emailId" id="emailId"/></td>
</tr>
<tr>
<td>Mobile Number:</td>
<td><sf:input path="mobileNumber" id="mNumber"/></td>
</tr>
<tr>
<td colspan="2" align="center">
<input type="submit" id="register" value="Register"/>
</td>
</tr>
</table>
</sf:form>
</center>
</body>
```

```
</html>
```

invalidEmail.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"

    pageEncoding="ISO-8859-1" isELIgnored="false"%>
<html>
<body bgcolor="lightyellow">
<h1>Swift Cart Portal</h1>

    <h2 id="register">Check the entered Email Id</h2>




</body>

</html>
```

InvalidMobileNumber.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1" isELIgnored="false"%>
<html>
<body bgcolor="lightyellow">
<h1>Swift Cart Portal</h1>

    <h2 id="register">Check the entered Mobile number</h2>
```

```
</body>


</html>
```

Success.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
        pageEncoding="ISO-8859-1" isELIgnored="false"%>
<html>
<body bgcolor="lightpink">
<h1>Swift Cart Portal</h1>
<h2 id="register">Customer registration is Successful for ${customer.name}</h2>


</body>
</html>
```